

Lecture Example 3

TNT (Pty) Ltd is an unlisted company^{+ risk}. On the valuation date (31 December 20X2) the following (summarized) forecast financial statements are provided:

Forecast statements of comprehensive income for the year ending 31 December:

	20X3	20X4	20X5
	Rm	Rm	Rm
Gross profit	188	247	289
Depreciation and amortization	(8)	(12)	(14)
Other operating expenses	(60)	(70)	(80)
Operating profit	120	165	195
Dividends received	2	2	3
Profit before interest	122	167	198
Finance costs	(6)	(6)	(7)
Profit before tax	116	161	191
Taxation	(32)	(45)	(53)
Profit for the year	84	116	138

Forecast statements of financial position as at 31 December:

	20X3	20X4	20X5
	Rm	Rm	Rm
ASSETS			
Non-current assets:			
Property, plant and equipment (carrying amount or NBV)	80	100	110
Other investments (at cost)	20	20	20
Current assets:			
Inventory	40	50	65
Accounts receivable	30	45	60
Cash and cash equivalents	7	17	15
Total assets	177	232	270
EQUITY AND LIABILITIES			
Equity:			
Total shareholders' equity	109	145	185
Non-current liabilities:			
Borrowings (at amortized cost)	58	67	65
Current liabilities:			
Accounts payable	6	14	13
Borrowings (current portion)	1	2	2
Tax payable	3	4	5
Total equity and liabilities	177	232	270

The following additional information is provided:

1. The WACC^{WACC} for the company is 18% and the income tax rate is 28%^T.
2. The carrying amounts of borrowings approximate their fair values.

3. The other investments represent a small shareholding of 2 million shares in a listed company. These shares had a closing price of 1 200 cents on 31 December 20X2. *R12*
4. All the cash and cash equivalents are considered as excess cash (that is, non-operating).
5. The carrying values as at 31 December 20X2 were as follows:

	Rm
Property, plant and equipment	50
Borrowings (non-current)	58
Borrowings (current)	1
Inventory	35
Trade receivables	25
Cash	5
Trade payables	4
Tax payable	1

6. The free cash flow *FCF* to the firm in 20X6 is expected to be 6% ⁹ higher than in 20X5 and is expected to grow at the same annual rate indefinitely.
7. The depreciation expense is assumed to be equal to the tax (wear and tear) allowances.

Required:

Calculate the fair value of a 75% ordinary shareholding in TNT (Pty) Ltd, using a valuation model based on free cash flow to the firm.

FCF

Lecture Example 3 – Suggested Solution

WORKINGS:

1. Calculating net investment in capital expenditure (for replacement and expansion):

	20X3	20X4	20X5
	Rm	Rm	Rm
Closing balance (CA of NCAs)	80	100	110
Add: Depreciation	8	12	14
	88	112	124
Less: Opening balance	(50)	(80)	(100)
Net investment in CAPEX	38	32	24

2. Calculating investment in net working capital (NWC):

	20X2	20X3	20X4	20X5
	Rm	Rm	Rm	Rm
Inventory	35	40	50	65
Accounts receivable	25	30	45	60
Accounts payable	(4)	(6)	(14)	(13)
Tax payable	(1)	(3)	(4)	(5)
Net working capital	55	61	77	107
Change in NWC		6	16	30

3. Calculating the FCFTF and the PV of FCFTF for the explicit forecast period and the CV:

	20X3	20X4	20X5
	Rm	Rm	Rm
NOPAT	86.4	118.8	140.4
Add: Depreciation	8	12	14
Gross operating cash flow	94.4	130.8	154.4
Less: net investment in CAPEX	(38)	(32)	(24)
Less: Net investment in WC	(6)	(16)	(30)
Free cash flow to the FIRM	50.4	82.8	100.4
CV $[100.4(1.06)]/(0.18 - 0.06)$			886.9
FCFTF with CV	50.4	82.8	987.3
PVIF at 18%	0.8475	0.7182	0.6086
PV of FCFTF with CV	42.7	59.5	600.9

4. Calculating the fair value of a 75% shareholding in TNT (Pty) Ltd:

	Rm
PV of operating cash flows (42.7 + 59.5 + 600.9)	703.1
Add: Non-operating assets:	
Fair value of other investments (2m shares x R12.00)	24.0
Cash and cash equivalents	5.0
Enterprise value	732.1
Less: Fair value of borrowings (58 +1)	(59.0)
Value of ordinary shares	673.1
 Fair value of a 75% shareholding (75% x R673.1m)	 504.8

POINTS TO NOTE:

1. The calculation of NOPAT given the profit for the year (after tax).

Using the 20X3 figures, the PROFIT AFTER TAX amount is adjusted for as follows:

	Rm
Profit after tax from continuing operations (excluding DOs)	84.0
LESS: Non-operating income:	
Dividend income (no tax effect)	(2.0)
Interest/rent income (net of tax) [Interest x (1 - t)]	n/a
ADD: Finance costs (after tax) [6 x (1 - 0.28)]	4.3
NOPAT (as previously calculated)	86.3*

*Rounding-up effect

2. Common Assumption: Depreciation is Equal to Tax (Wear and Tear) Allowances:

It is common to assume that depreciation expense is equal to tax (wear and tear) allowance. This is meant to simplify the calculation of NOPAT using the EBITABNOI (earnings before interest and tax and before non-operating income). In this case, NOPAT is calculated simply as $EBITABNOI \times (1 - t)$. This is because EBITABNOI is equal to taxable income. Where the depreciation expense is different from tax (wear and tear) allowance, NOPAT is calculated as follows (assuming wear and tear allowance of R12.0m for 20X3):

	Rm
EBIT and before non-operating income	120.0
Less: Tax (see working below)	(32.5)
NOPAT	87.5
Add: Depreciation	8.0
Gross operating cash flow	95.5

Working – Tax calculation:

	Rm
EBIT and before non-operating income	120.0
Add: Depreciation	8.0
Less: Wear and tear allowance	(12.0)
Taxable income	116.0
Tax expense (28% x R116.0)	<u>32.48</u>

Note that the assumption (that depreciation expense is equal to wear and tear allowance) has no implication on how you calculate NOPAT using the profit for the year (after tax) amount. This is because any difference between depreciation and tax allowance would have been taken into account when calculating the tax expense for each year of the explicit forecast period.

3. Valuation model based on FREE CASH FLOW TO EQUITY:

- Of the two models based on free cash flows, the model based on free cash flow to equity (FCFTE) is less popular; hence it is not discussed in detail in this module.
- The two models (FCFTF and FCFTE) differ mainly with regard to how free cash flows are calculated and the discount rate used.
- Free cash flow to equity is determined after deducting interest expense and preference share dividends (that is, free to distribute to ordinary shareholders).
- The free cash flow to equity (including the CV) is discounted using the COST OF EQUITY and **not** WACC.